

Finley Bate-Smith

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Education

Queen's University | Kingston, ON

Graduation date: April 2028

- BS Mechanical Engineering – GPA: 3.5
- Dean's Honours List: 2025-2026

Experience

Royal Military College of Canada | Kingston, ON

May 2026 - Present

Undergraduate Research Assistant

- Awarded Undergraduate Student Research Award for supporting summer research under Dr. Lau-Chapdelaine
- Developed control system for a two-stage light gas gun using Arduino and Python/tkinter, enabling remote valve actuation and real-time monitoring of 6 pressure transducers
- Implemented adjustable-rate data logging, live pressure plotting and emergency valve-shutdown safeguards to the graphical user interface for high-pressure operation
- Designed an induction-coil velocimeter, using Python to optimize coil geometry and SolidWorks to model the coil spool, validating results within 1.6% of a Garmin chronograph
- Designed a light-gate velocimeter, using IR emitter-detector pairs and 3D-printed mounting brackets, achieving 3% repeatability and validating results within 8% against a Garmin chronograph

VCMT | Vancouver, BC

May 2025 – August 2025

Internship Coordinator

- Scheduled 240 registered massage therapy internship students across 15 outreach locations and a student clinic with over 800 appointments per week
- Problem-solving daily by managing absences, reallocating students from one location to another, scheduling make-up shifts, and communicating with students about changes in their schedule
- Tracked student internship metrics such as the number of patients and treatment hours to determine if graduation requirements are met
- Prepared reports for management detailing student internship metrics and program scheduling efficiency

Queen's Formula SAE Design Team | Kingston, ON

September 2024 – Present

Aerodynamics Team Member

- Modeled the Q26 rear wing swan-neck mounts in SolidWorks and the supporting rear wing mounting rib in CATIA to secure the rear wing assembly on the chassis, complying with FSAE rulebook
- Designed the Q26 car's sidepod in CATIA after relocation of the radiator, developing a new aerodynamic profile optimized for downforce and reducing drag, while complying with FSAE rulebook
- Collaborated weekly in engineering design reviews with the team lead and sub-teams to support the development of an EV race car to compete in New Hampshire and Michigan

Proscape Landscape | Vancouver, BC

June 2022 – August 2024

Landscaper

- Maintained lawns, gardens, and outdoor spaces by performing tasks such as mowing, planting, and weeding, ensuring visually appealing and healthy landscapes
- Operated and maintained equipment, including lawnmowers, blowers, trimmers, and rakes to complete tasks efficiently and safely
- Collaborated with team members to execute landscaping projects on schedule and within client specifications

Skills

- **Software:** Solidworks, CATIA V5, ANSA, Arduino
- **Manufacturing:** 3D Printing, MIG Welding, Manual Machining
- **Programming:** C, C++, Python